

To

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Editors-in-Chief
Science Advances

Dear Editors,

We are delighted to submit our manuscript entitled "Heatwave risk reshaped by urban expansion-clustering nexus" for consideration in *Science Advances*. This work has been approved by all co-authors, and we declare no competing interests. The manuscript has not been published or presented elsewhere and is not under consideration by another journal.

As global urbanization merges discrete cities into interconnected megalopolises, adjacent urban heat domes (UHDs) increasingly coalesce. Yet, conventional climate adaptation still treats cities as isolated thermal entities, neglecting the transboundary atmospheric dynamics triggered by this spatial integration. **While the localized thermal impacts of individual city expansion have been widely studied, the regional-scale atmospheric dynamics that emerge when expanding UHDs physically merge remain underexplored.**

As urban agglomerations continue to grow under a warming climate, understanding how this expansion-clustering nexus reshapes multidimensional heatwave risks is crucial for sustainable regional planning and climate adaptation. In this study, by combining a causal machine learning framework with an observational dataset of 1,690 Chinese cities over three decades (1990–2022), we decouple urbanization-induced thermal penalties from macroscale climate variability. We show that:

- (1) While macroscale climate warming largely dictates the absolute baselines of peak heatwave intensity and seasonal onset, **urban spatial expansion primarily drives the regional escalation of heatwave frequency and duration.**
- (2) Within this urbanization-induced thermal penalty (ΔHW), a distinct structural dichotomy emerges: **internal 3D densification elevates localized peak temperatures (ΔHWA) and advances seasonal onset (ΔHWO)** via radiation trapping, whereas **external intercity coalescence escalates regional heatwave frequency (ΔHWF) and duration (ΔHWD)** by increasing aerodynamic resistance.
- (3) Regional heat dome coalescence is dictated by explicit spatial thresholds. **When intercity gaps contract below ~50 km, adjacent UHDs structurally merge into a contiguous 'mega-UHD' that homogenizes regional thermal fields, while extreme proximity (<25 km) triggers a secondary surge in heatwave frequency** due to severe aerodynamic stagnation.

These findings highlight the need to transition from isolated municipal interventions to transboundary regional planning that preserves intercity spatial buffers and optimizes the topological connectivity of blue-green infrastructure.

We believe this work meets the criteria of *Science Advances* for broad interdisciplinary impact by providing the following contributions:

- (1) **A transboundary shift in the adaptation paradigm:** Challenging the isolated-city model, we demonstrate that structural coalescence transforms extreme heat from a

localized intra-urban issue into a systemic, transboundary regional hazard.

- (2) **Large-scale methodological rigor:** By applying dual-validation causal inference across 1,690 cities over 33 years, we successfully decouple urban morphological and spatial forcings from macro-climatic noise, linking empirical data directly to physical mechanisms.
- (3) **Actionable physical thresholds for regional planning:** Moving beyond statistical correlations, we quantify explicit spatial limits (e.g., the ~50-km mega-UHD trigger) and prove that optimizing blue-green infrastructure connectivity structurally outperforms increasing absolute coverage.

Thank you for your time and consideration of our manuscript. We look forward to your feedback and suggestions.

With kind regards,



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